

Experiment 19 — Naive Bayes Classification for Bank Loan Prediction

AIM

To implement Bank Loan Prediction using the Naive Bayes classification algorithm in Google Colab.

SOFTWARE USED

Google Colab

PROCEDURE

1. Open Google Colab and create a new notebook.
2. Import the required libraries.
3. Create a sample bank loan dataset containing applicant details.
4. Separate the input features and loan approval status.
5. Split the dataset into training and testing data.
6. Create a Gaussian Naive Bayes classifier.
7. Train the model using the training data.
8. Predict whether the loan is approved or rejected.
9. Calculate the classification accuracy.
10. Test the model with a new loan applicant.

CODE

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score

# Create Bank Loan dataset

data = {
    "Income": [25000, 30000, 45000, 50000, 60000,
```

```

        20000, 35000, 70000, 80000, 28000,
        40000, 55000, 65000, 22000, 75000,
        32000, 48000, 90000, 27000, 58000],
    "Age": [25, 30, 35, 40, 45,
            23, 32, 50, 55, 28,
            38, 42, 48, 24, 52,
            31, 37, 60, 26, 44],
    "LoanAmount": [100000, 150000, 200000, 250000, 300000,
                   80000, 180000, 350000, 400000, 120000,
                   220000, 270000, 320000, 90000, 380000,
                   160000, 210000, 450000, 110000, 280000],
    "CreditScore": [650, 680, 720, 750, 780,
                    620, 700, 800, 820, 640,
                    710, 740, 770, 610, 810,
                    690, 730, 850, 630, 760],
    "Loan_Status": [0, 0, 1, 1, 1,
                    0, 1, 1, 1, 0,
                    1, 1, 1, 0, 1,
                    0, 1, 1, 0, 1]
}

# Create DataFrame
df = pd.DataFrame(data)
print("Bank Loan Dataset")
print(df.head())

# Separate features and target
X = df.drop("Loan_Status", axis=1)
y = df["Loan_Status"]

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(

```

```
X,  
y,  
test_size=0.2,  
random_state=42,  
stratify=y  
)  
# Create Naive Bayes model  
model = GaussianNB()  
# Train model  
model.fit(X_train, y_train)  
# Predict  
y_pred = model.predict(X_test)  
# Calculate accuracy  
accuracy = accuracy_score(y_test, y_pred)  
print("\nAccuracy:", accuracy)  
# Predict a new loan applicant  
new_applicant = [[50000, 35, 200000, 730]]  
prediction = model.predict(new_applicant)  
if prediction[0] == 1:  
    print("New Applicant: Loan Approved")  
else:  
    print("New Applicant: Loan Rejected")
```

OUTPUT

Bank Loan Dataset

	Income	Age	LoanAmount	CreditScore	Loan_Status
0	25000	25	100000	650	0
1	30000	30	150000	680	0
2	45000	35	200000	720	1
3	50000	40	250000	750	1
4	60000	45	300000	780	1

Accuracy: 1.0

RESULT

Thus, Naive Bayes classification for Bank Loan Prediction was successfully implemented in Google Colab, and the loan approval status of applicants was predicted successfully.